Forecasting with stationary models IV

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Plan for this slide deck

1. Moving average model
2. ARMA model
3. Convenient technical tool: lag polynomials
   - Stationarity of AR(p) model
4. AR vs. ARMA

Moving average model

- In an AR model, current outcomes depend on lagged outcomes.
- Different way to model time series dependence: lagged innovations instead of lagged outcomes.

$$y_t = \mu + u_t + \theta u_{t-1},$$

where u_t is a white noise process with variance σ_u^2: $u_t \sim WN(\sigma_u^2)$.

- Called a **moving average** (MA).
- We can generalize to q lagged innovations: MA(q).

$$y_t = \mu + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}.$$

- Always cov. stationary, regardless of the parameters $\theta_1, \ldots, \theta_q, \sigma_u^2$.

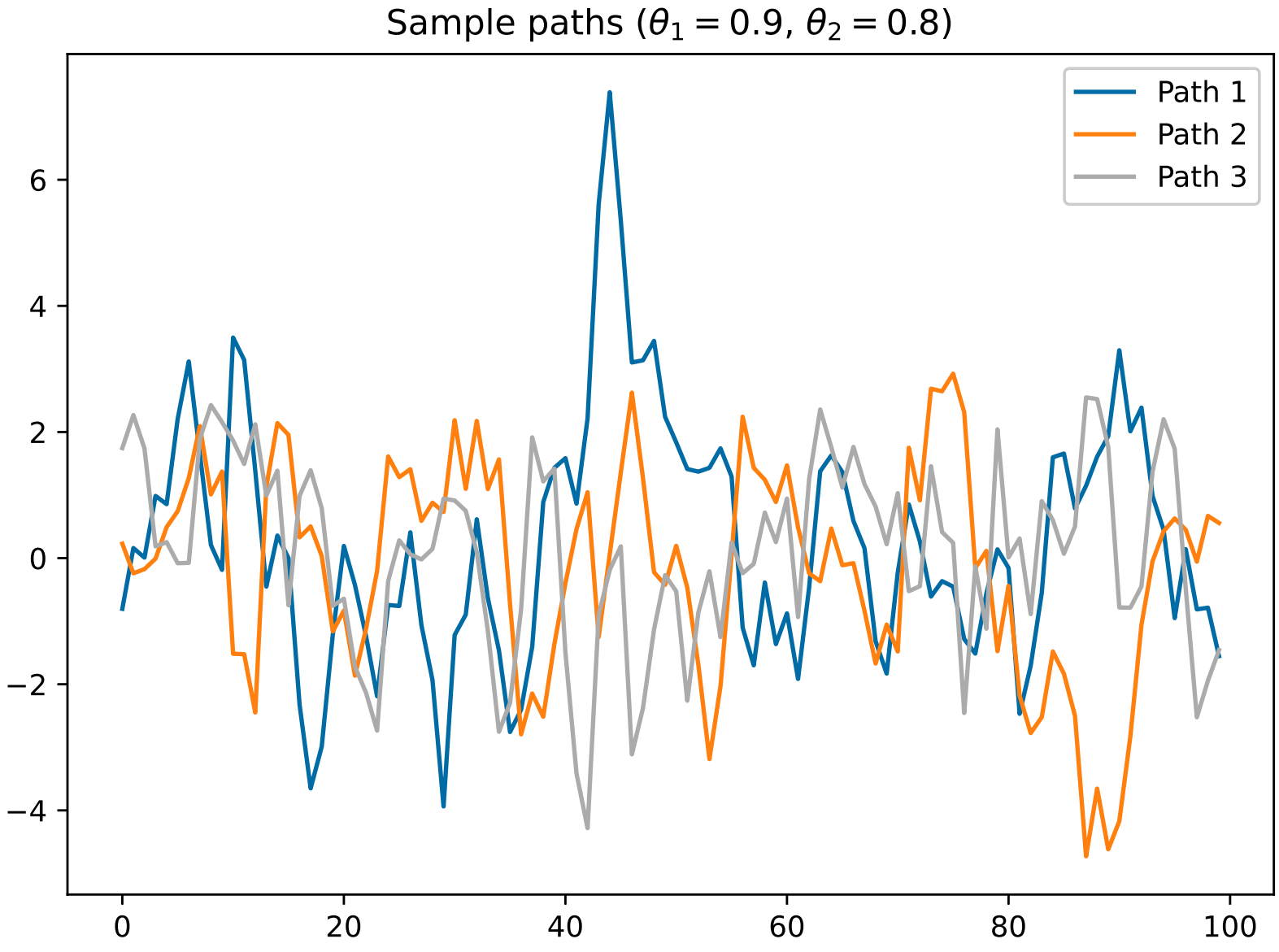
MA(q) model: ACF

$$y_t = \mu + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}, \quad u_t \sim WN(\sigma_u^2)$$

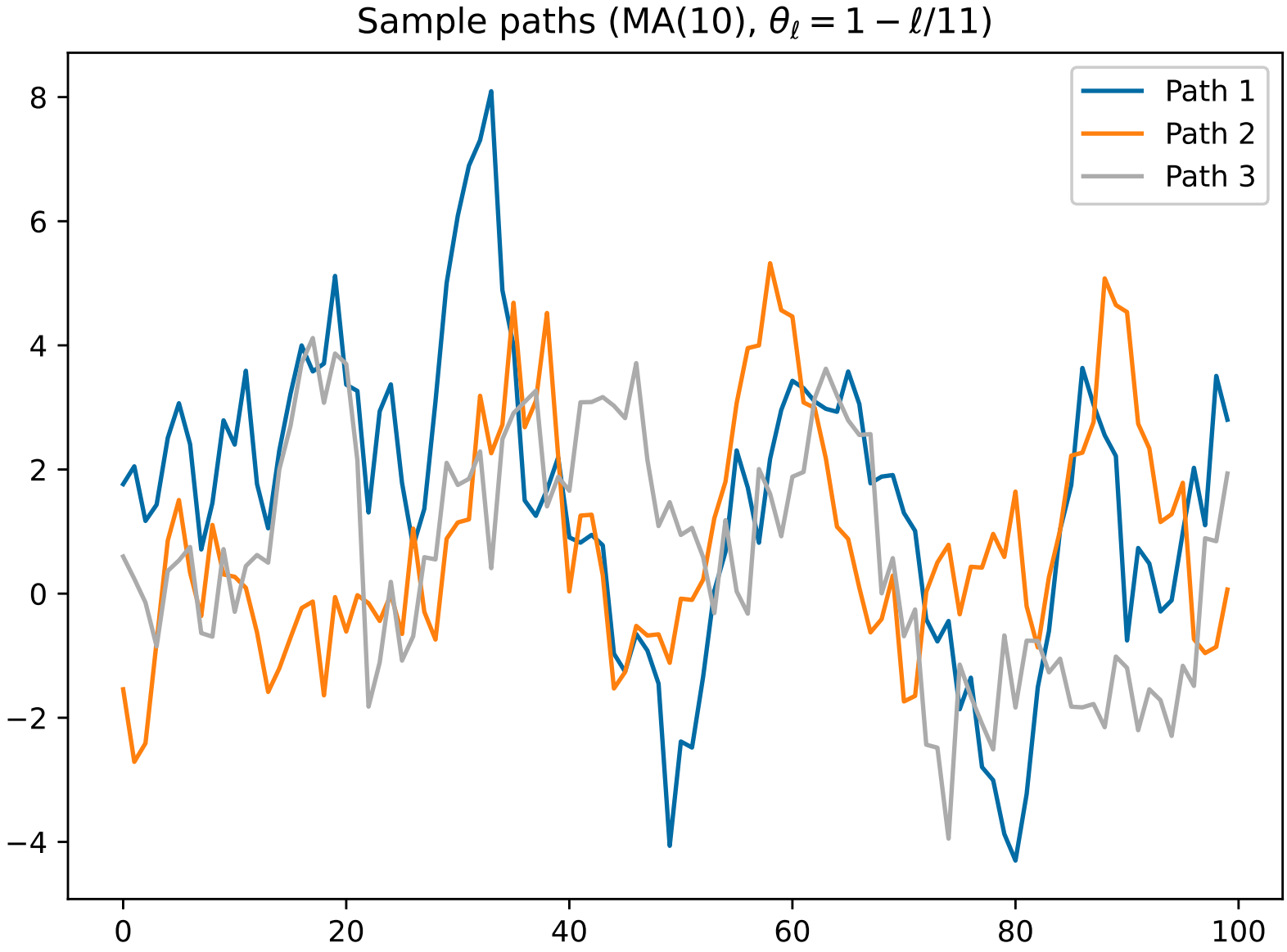
- Clearly, $E(y_t) = \mu$.
- Exercise: show

$$\gamma(\ell) \equiv \text{Cov}(y_t, y_{t-\ell}) = \begin{cases} \sigma_u^2 \sum_{j=0}^{q-\ell} \theta_j \theta_{j+\ell} & \text{for } 0 \leq \ell \leq q, \\ 0 & \text{for } \ell > q, \end{cases}$$

  where $\theta_0 \equiv 1$.
- The process exhibits **finite dependence**: autocorrelations are exactly zero after q lags (unlike AR processes).



$$y_t = u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2}, \quad u_t \overset{iid}{\sim} N(0,1), \quad \theta_1 = 0.9, \theta_2 = 0.8$$



$$y_t = u_t + \sum_{\ell=1}^{10} \theta_\ell u_{t-\ell}, \quad u_t \overset{iid}{\sim} N(0,1), \quad \theta_\ell = 1 - \frac{\ell}{11}$$

Wold decomposition

- MA processes turn out to be very general.
- Result (**Wold decomposition**): any cov. stationary time series $\{y_t\}$ can be arbitrarily well approximated as $q \to \infty$ by

$$y_t = \zeta_t + u_t + \sum_{\ell=1}^{q} \theta_\ell u_{t-\ell}, \quad \text{where:}$$

  - u_t is some white noise process.
  - ζ_t is a perfectly predictable process (typically just a constant mean).
- If the autocovariances of $\{y_t\}$ satisfy $\gamma(\ell) = 0$ for all $\ell > m$ (finite dependence), then y_t can be represented as an MA(m) process (plus perfectly predictable part).
- But if the ACF decays slowly to zero, then we may need large q to get a good approximation.

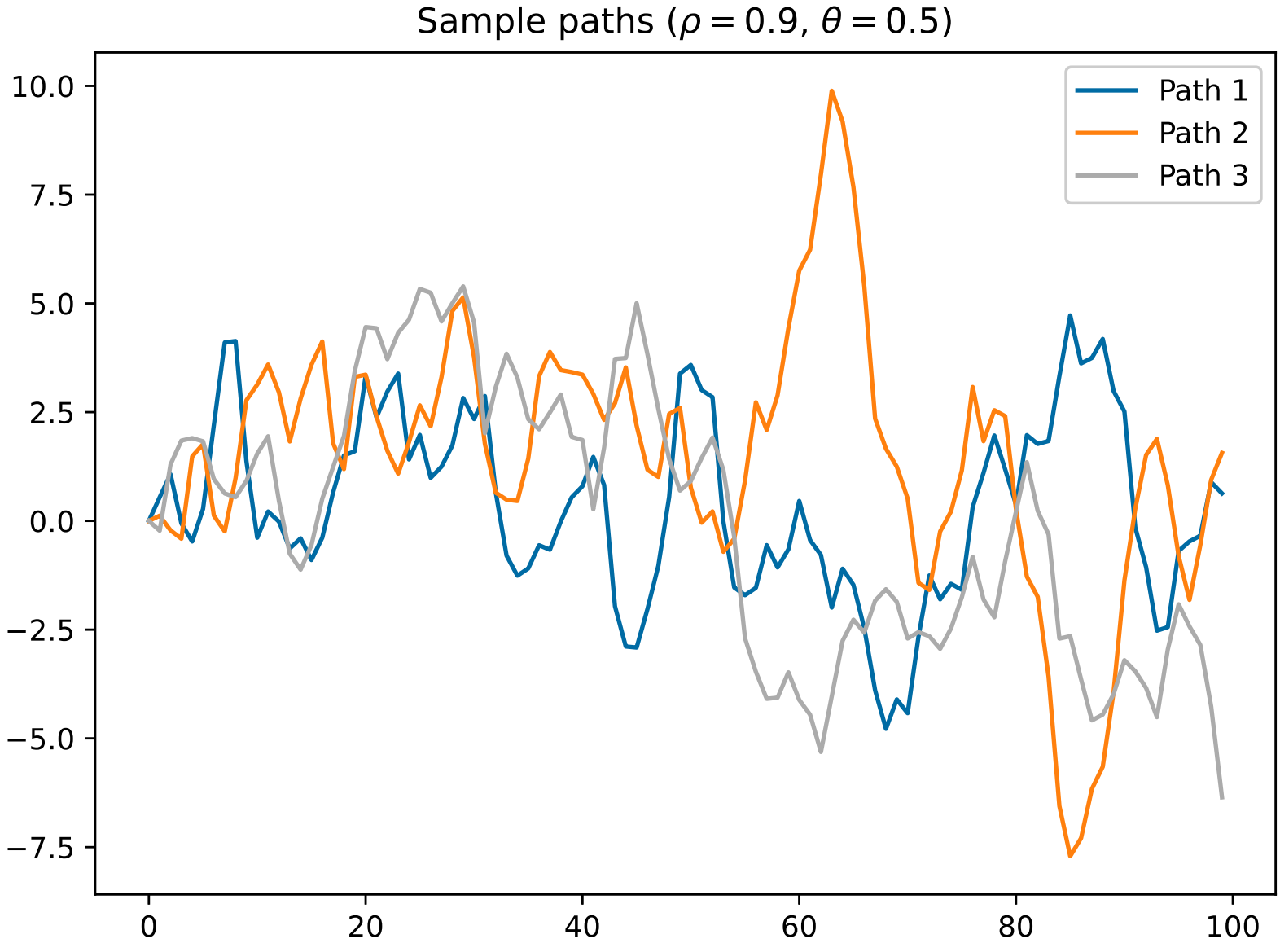
ARMA(p, q) model

- We can combine the AR and MA modeling frameworks:

$$y_t = c + a_1 y_{t-1} + \cdots + a_p y_{t-p} + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q},$$

$$u_t \sim WN(\sigma_u^2).$$

- Called an **ARMA(p, q) model**.
  - p AR lags.
  - q MA lags.
- Rich modeling framework for capturing stationary time series dynamics.



$$y_t = \rho y_{t-1} + u_t + \theta u_{t-1}, \quad u_t \overset{iid}{\sim} N(0,1), \quad \rho = 0.9,\ \theta = 0.5$$

Loose ends

- We have some loose ends:
  1. When is an AR/ARMA process covariance stationary?
  2. How do we forecast an ARMA(p, q) process?
  3. How to estimate ARMA parameters?
- To make progress on #1–2, it's helpful to work with lag polynomials (next slides).
- We will tackle #3 much later when we talk about state space models.

Lag operator

- It's convenient to use the **lag operator** notation

$$Ly_t \equiv y_{t-1}.$$

  The operator L replaces a time series $\{y_t\}$ with its lag $\{y_{t-1}\}$.

- Applying it twice, we get the second lag:

$$L^2 y_t = L(Ly_t) = Ly_{t-1} = y_{t-2}.$$

- And so on:

$$L^\ell y_t = y_{t-\ell}.$$

- Negative powers give leads:

$$L^{-\ell} y_t = y_{t+\ell}.$$

- And L^0 is the identity operator:

$$L^0 y_t = y_t.$$

Lag polynomial

- We can now define a p-th order **lag polynomial**:

$$b(L) = b_0 + b_1 L + \cdots + b_p L^p = \sum_{\ell=0}^{p} b_\ell L^\ell.$$

- This just means that

$$b(L)y_t = b_0 y_t + b_1 y_{t-1} + \cdots + b_p y_{t-p} = \sum_{\ell=0}^{p} b_\ell y_{t-\ell}.$$

- We also call a lag polynomial a **linear filter**.
- If $\{y_t\}$ is covariance stationary, then so is $b(L)y_t$.

Two-sided linear filters

- We can generalize to a **two-sided** linear filter:

$$b(L) = \sum_{\ell=-p}^{p} b_\ell L^\ell \quad \Longrightarrow \quad b(L)y_t = \sum_{\ell=-p}^{p} b_\ell y_{t-\ell} \quad \text{(lags and leads)}.$$

  (If $b_\ell = 0$ for $\ell < 0$, the filter is **one-sided**.)

- Examples of linear filters:
  - First difference: $b(L)y_t = \Delta y_t \equiv y_t - y_{t-1}$ $(b_0 = 1, b_1 = -1)$.
  - One-year change (monthly data): $b(L)y_t = y_t - y_{t-12}$ $(b_0 = 1, b_1 = \cdots = b_{11} = 0, b_{12} = -1)$.
  - Three-period central moving average: $b(L)y_t = (y_{t-1} + y_t + y_{t+1})/3$ $(b_{-1} = b_0 = b_1 = 1/3)$.

Multiplication of lag polynomials

- Start with a time series that's a linear filter:

$$y_t = b(L)u_t, \quad \text{with} \quad b(L) = \sum_{\ell=-p}^{p} b_\ell L^\ell.$$

- Now take a linear filter $d(L) = \sum_{\ell=-q}^{q} d_\ell L^\ell$ of that series:

$$z_t = d(L)y_t = g(L)u_t, \quad \text{where} \quad g(L) \equiv d(L)b(L).$$

- $g(L) = \sum_{\ell=-(p+q)}^{p+q} g_\ell L^\ell$ is a linear filter of order $p + q$.
  - Exercise: what are its coefficients g_ℓ?
- Facts:
  1. $b(L)d(L) = d(L)b(L)$.
  2. If d is a constant, $b(L)d = b(1)d = d\sum_{\ell=-p}^{p} b_\ell$.

Example: multiplication of lag polynomials

$$y_t \equiv \Delta u_t = (1 - L)u_t$$

$$z_t \equiv \Delta y_t = (1 - L)y_t = (1 - L)^2 u_t$$

$$(1 - L)^2 = 1 - 2L + L^2$$

$$\Downarrow$$

$$z_t = u_t - 2u_{t-1} + u_{t-2}$$

AR(p): towards stationarity

- Now let's use lag polynomials to analyze the AR(p) model:

$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t.$$

- Using our new notation, we can write this compactly:

$$a(L) y_t = c + u_t, \quad \text{where} \quad a(L) \equiv 1 - \sum_{\ell=1}^{p} a_\ell L^\ell.$$

- *Suppose* we could find a one-sided lag polynomial $b(L)$ such that $b(L)a(L) = 1$. Then

$$y_t = b(L)a(L)y_t = b(L)(c + u_t).$$

$\Longrightarrow$ y_t is a one-sided linear filter of the white noise process u_t
$\Longrightarrow$ y_t is cov. stationary.

Inverse of lag polynomial

- A one-sided lag polynomial $b(L)$ satisfying $b(L)a(L) = 1$ is called the **inverse** of $a(L)$, written $b(L) = a(L)^{-1}$.
- Result: $a(L) = 1 - \sum_{\ell=1}^{p} a_\ell L^\ell$ has a one-sided inverse, $a(L)^{-1}$, if all the complex solutions of the **characteristic equation**

$$a(z) = 0$$

lie outside the unit circle, i.e., $|z| > 1$.

- Writing out the characteristic equation explicitly:

$$a_1 z + a_2 z^2 + \cdots + a_p z^p = 1.$$

This has exactly p complex solutions (fundamental theorem of algebra). If they're all larger than 1 in modulus, then $a(L)^{-1}$ exists.

Technical detail: infinite-order lag polynomials

- The inverse $b(L) = \sum_{\ell=0}^{\infty} b_\ell L^\ell = a(L)^{-1}$ might be of **infinite order**, i.e., $b_\ell \neq 0$ even for large ℓ.
- Not a mathematical problem: everything we've said goes through when the lag polynomial has ∞-ly many terms.
- There is one requirement though: for an infinite-order polynomial $b(L) = \sum_{\ell=0}^{\infty} b_\ell L^\ell$ to yield a mathematically well-defined process $b(L)y_t$ for any cov. stationary $\{y_t\}$, its coefficients must be **absolutely summable**:

$$\sum_{\ell=0}^{\infty} |b_\ell| < \infty.$$

- Thankfully, if the characteristic equation $a(z) = 0$ has all solutions outside the unit circle, then there does indeed exist a one-sided $a(L)^{-1}$ with absolutely summable coefficients.

AR(p): stationary condition

- To summarize, we've seen that the AR(p) model

$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t, \quad u_t \sim WN(\sigma_u^2),$$

has a cov. stationary solution $\{y_t\}$ when the characteristic equation

$$a(z) = 1 - \sum_{\ell=1}^{p} a_\ell z^\ell = 0$$

has all its solutions z (called **roots**) outside the unit circle.

- Moreover, in this case,

$$y_t = a(L)^{-1}(c + u_t) = \frac{c}{1 - \sum_{\ell=1}^{p} a_\ell} + a(L)^{-1} u_t,$$

so $\{y_t\}$ is also an ∞-order MA process.

Special case: AR(1)

$$y_t = c + \rho y_{t-1} + u_t$$

- Characteristic equation:

$$\rho z = 1 \quad \Longrightarrow \quad z = \frac{1}{\rho}.$$

  So $|z| > 1$ if and only if $|\rho| < 1$. Same as we derived previously!

- Inverse of AR polynomial:

$$(1 - \rho L)^{-1} = \sum_{\ell=0}^{\infty} (\rho L)^{\ell} \quad \text{(geometric series)}.$$

- So y_t has the MA representation

$$y_t = (1 - \rho L)^{-1}(c + u_t) = \boxed{\frac{c}{1-\rho} + \sum_{\ell=0}^{\infty} \rho^{\ell} u_{t-\ell}}.$$

Special case: AR(1) – continued

- We could also have obtained the MA representation by repeated substitution in the AR equation:

$$
\begin{aligned}
y_t &= c + \rho y_{t-1} + u_t \\
&= c + \rho(c + \rho y_{t-2} + u_{t-1}) + u_t \\
&= c + \rho(c + \rho(c + \rho y_{t-3} + u_{t-2}) + u_{t-1}) + u_t \\
&\vdots
\end{aligned}
$$

- . . . but the lag polynomial method is much simpler and more generalizable when you get used to the mechanics.

ARMA models: stationarity and invertibility

- Write an ARMA(p, q) model compactly using lag polynomials:

$$a(L)y_t = c + \theta(L)u_t, \quad a(L) \equiv 1 - \sum_{\ell=1}^{p} a_\ell L^\ell, \quad \theta(L) \equiv 1 + \sum_{\ell=1}^{q} \theta_\ell L^\ell.$$

- If $a(L)$ is invertible, then

$$y_t = \mu + d(L)u_t, \quad \text{where} \quad d(L) \equiv a(L)^{-1}\theta(L), \quad \mu \equiv a(1)^{-1}c,$$

  so y_t is a (cov. stationary) MA(∞) process.

- If $\theta(L)$ is invertible, then

$$u_t = \psi(L)(y_t - \mu), \quad \text{where} \quad \psi(L) \equiv \theta(L)^{-1}a(L),$$

  and u_t is **invertible**: function of current and lagged $y_t, y_{t-1}, y_{t-2}, \ldots$, but not of future $y_{t+1}, y_{t+2}, \ldots$.

ARMA models: forecasting

$$y_t = c + \sum_{\ell=1}^{p} a_\ell y_{t-\ell} + u_t + \sum_{\ell=1}^{q} \theta_\ell u_{t-\ell}, \quad E[u_t \mid \{y_s\}_{s<t}] = 0$$

- Assume the ARMA model is cov. stationary and invertible. Then

$$u_{t-\ell} = \psi(L)(y_{t-\ell} - \mu) \quad \text{(function of } y_{t-\ell}, y_{t-\ell-1}, \dots\text{)}.$$

- Best one-step-ahead forecast:

$$E[y_t \mid y_{t-1}, y_{t-2}, \dots] = c + \sum_{\ell=1}^{p} a_\ell y_{t-\ell} + \sum_{\ell=1}^{q} \theta_\ell \psi(L)(y_{t-\ell} - \mu).$$

- In general, this depends on ∞-ly many lags $y_{t-1}, y_{t-2}, \dots$, not just a finite number p like for AR(p) models.

Forecasting example: MA(1)

$$y_t = u_t + \theta u_{t-1} = \theta(L)u_t, \quad \theta(L) \equiv 1 + \theta L$$

- If $|\theta| < 1$, then

$$\theta(L)^{-1} = \sum_{\ell=0}^{\infty}(-\theta L)^{\ell} \quad \text{(geometric series)}$$

$$\Downarrow$$

$$u_t = \theta(L)^{-1}y_t = \sum_{\ell=0}^{\infty}(-\theta)^{\ell}y_{t-\ell}$$

$$\Downarrow$$

$$E[y_t \mid y_{t-1}, y_{t-2}, \ldots] = \theta \times E[u_{t-1} \mid y_{t-1}, y_{t-2}, \ldots]$$

$$= \theta \times \theta(L)^{-1}y_{t-1} = \boxed{\theta y_{t-1} - \theta^2 y_{t-2} + \theta^3 y_{t-3} - \ldots}$$

AR or ARMA?

- ARMA models can capture complicated dynamics, but difficult to estimate/forecast (we'll talk about estimation later).
- AR models easy to estimate and forecast.
- Any stationary+invertible ARMA(p, q) implies an AR(∞) model:

$$\psi(L)(y_t - \mu) = u_t, \quad \psi(L) \equiv \theta(L)^{-1} a(L).$$

  So if we control for enough lags $\tilde{p}$ in an AR$(\tilde{p})$ regression, we can capture nearly the same dynamics as an ARMA(p, q).

  - May require $\tilde{p}$ to be much larger than the original p, q in some cases.
- In practice, AR modeling is much more popular than ARMA.

What did we learn?

- MA model
- ARMA model
- Lag polynomials
  - Connection to linear filters
  - Multiplication
  - Inverse
- Condition for cov. stationarity of AR model
- Forecasting with ARMA model

Looking ahead

- How can we analyze the relationships between multiple time series?
- How can we use multiple time series to forecast?